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Data Analysis Course

The Analysis of the Performance of Data Analysis Students

Submitted by

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The Impact of Artificial Intelligence Tools on Students' Learning and Academic Performance

Introduction

Artificial Intelligence (AI) has become an integral part of modern education, with students increasingly relying on tools such as ChatGPT and other AI-based platforms. These tools provide quick access to information, assist in completing assignments, and enhance the overall learning experience.

However, despite these advantages, there is growing concern that excessive reliance on AI may reduce students' independent thinking, problem-solving abilities, and academic effort.

This study aims to examine the impact of AI usage on students' learning behaviour and academic performance, highlighting both its benefits and potential drawbacks.

Research Question

Does the use of artificial intelligence tools affect students' learning and academic performance positively or negatively?

Hypothesis

Null Hypothesis (H_0): The use of AI tools has no significant effect on students' academic performance.

Alternative Hypothesis (H_1): The use of AI tools significantly affects students' academic performance, either positively or negatively.

Population of Interest:

The population of interest consists of **university students who use AI tools** in their academic activities. This includes students from various disciplines, academic levels, and institutions.

Sampling Method:

The sampling method used in this study is **convenience sampling**. The survey was distributed online to university students who actively use AI tools.

This method was chosen because it:

1 - Provides easy access to participants

2 -Allows for quick data collection

3 -Is cost-effective and practical

Although convenient, this method may limit the generalizability of the results.

Bias Identification:

Several potential sources of bias were considered in this study:

1- **Self-report bias:** Participants may not accurately report their AI usage
2- **Selection bias:** Only students with access to the survey participated
3- **Interpretation bias:** Differences in understanding AI tools among students

To minimize bias:

1- Questions were designed to be **neutral and non-leading**

2- Responses were collected **anonymously**
3- Clear and simple language was used to avoid confusion

Despite these efforts, some bias may still exist.

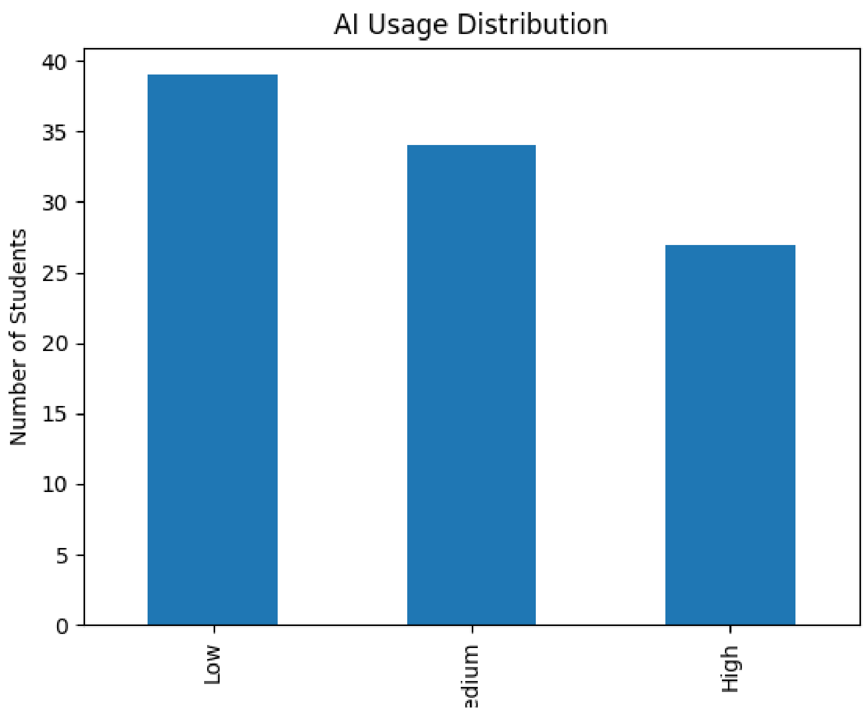
Survey Questions:

Survey Questions

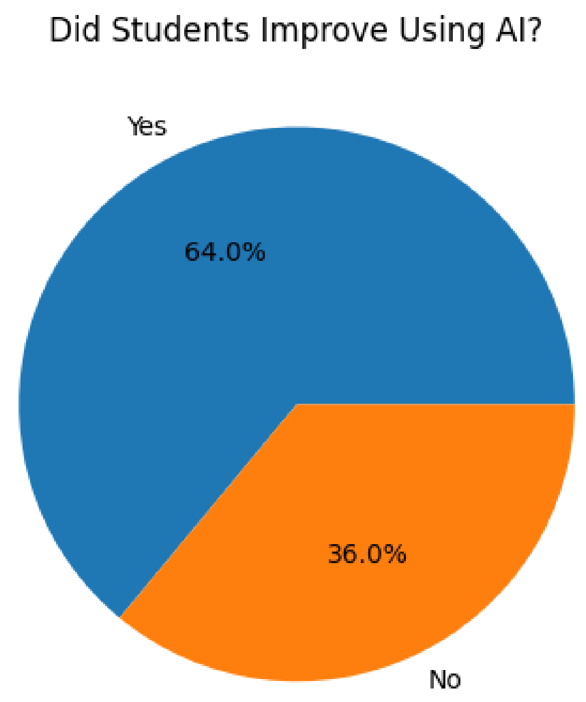
1. How often do you use AI tools in your studies?
2. Which AI tools do you use most frequently?
3. For what purposes do you use AI tools (e.g., assignments, studying, research)?
4. Do AI tools help you understand academic concepts better?
5. Have your grades improved since using AI tools?
6. To what extent do you rely on AI tools to complete assignments?
7. Do you believe AI tools reduce your independent thinking skills?
8. How much time do AI tools save you on average?
9. Do you verify the accuracy of AI-generated information?
10. Do you think AI tools should be regulated in education?

Online survey link: We generated information from social media websites.
Number of samples collected: 100

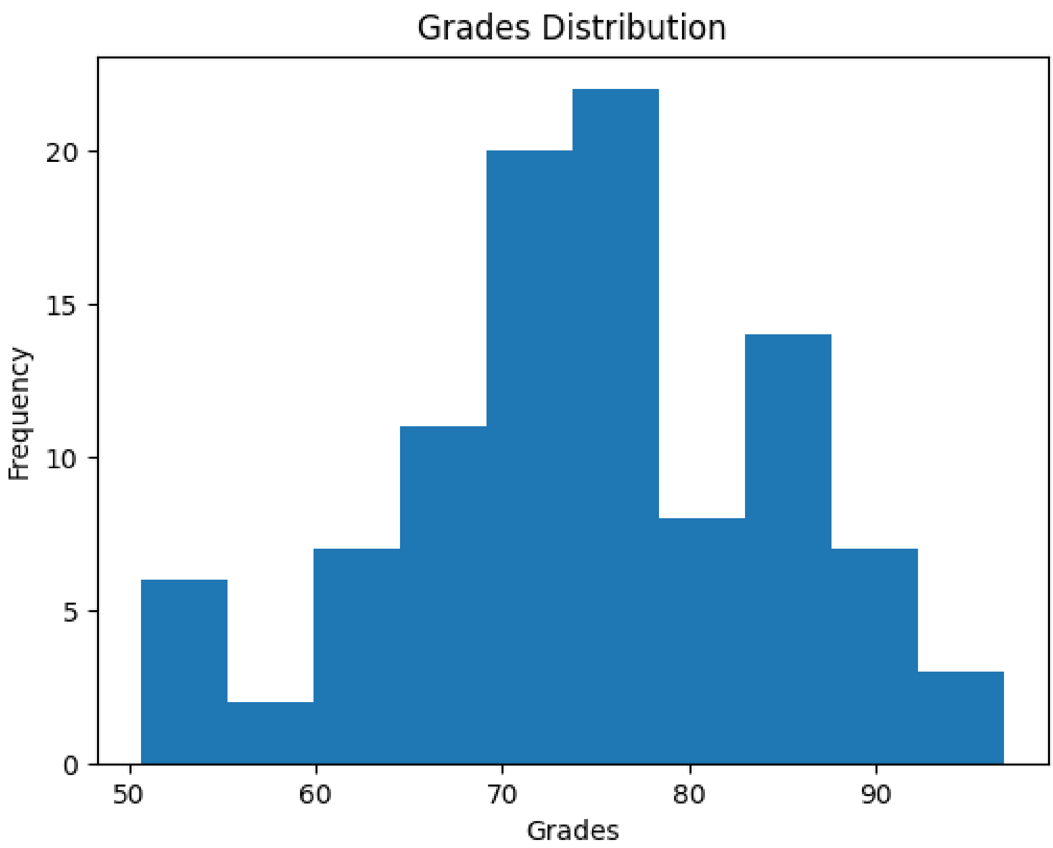
Analysis



This bar chart illustrates the frequency of AI tool usage among students. It can be observed that most students fall within the medium to high usage categories. This suggests that AI tools are widely adopted in academic environments.



The pie chart shows the proportion of students who believe their performance improved after using AI tools. A majority of students reported improvement, indicating a generally positive perception of AI's impact on learning.



The histogram displays the distribution of students' grades. Most grades are concentrated around the average range, with fewer students achieving very high or very low scores. This suggests a relatively normal distribution of academic performance.

Conclusion

Based on the analysis, artificial intelligence tools appear to have both **positive and negative effects** on students' learning:

Positive effects:

1- Faster understanding of complex topics

2- Increased productivity and efficiency

Negative effects:

1- Reduced critical thinking skills

2- Over-reliance on automated solutions

Therefore, AI tools should be used as **supportive learning aids** rather than replacements for independent thinking and effort.

Any potential issues

This study has several limitations:

1- The data is based on **self-reported responses**, which may not always be accurate

2- The sample size is relatively small and may not represent all students

3- Convenience sampling limits the generalizability of the findings

Future research should:

1- Include a larger and more diverse sample

2- Use experimental or longitudinal methods

3- Compare results across different academic fields
